1. Battery in and E-stop bank B (LM5069-1, two FETs). Bank A, in series ahead of it, is block 7

Pack: 8 AWG pigtail pair to an XT90 (mates the pack XT90S), 1 BAT, 2 GND

The bank is the soft start for all the motor capacitance BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Console is not behind it.
Startup: C4 (100 nF) sets dV/dt ≈ 0.1 V/ms; with R4 (374 Ω) and R5 (22 kΩ) the fault trip is ~0.5 A, so the load must draw under ~0.5 A while C4 charges.
Fault: Cdv (10) nF sets 22×2.5 µA, 100×225 A/s; 0.25×0.5 A to fold. Also setting 0.5 mF at 2.06 A for 4.2 ms; 0.6 A for 2.5 ms; 0.8 A many. Give a cap unpopulated.
Power limit 652 W at 26.5×6.9, 480–832 W with tolerance: RPWR = 1.5×5 x Rs × IF − 1.18 mV x VDS / Rs). SNVSV52C eq 10. VDAG 0 mV over the 5 mV floor. Timer 4.7 nF, 0.22 ms typ, 0.15–0.39 ms. Retry 1.5 s off, 0.1 s on, 10 tries, then latch. Set R6 = 33 kΩ for 0.1 s on, 1.5 s off, 10 tries, then latch. Set R6 = 22.7 k at 0.59 ms 1.4 s. Read the real curve at layout.
FETs: 0.85 mOhm for the pair, 0.2 W hot at 60 A, 1.3 W at 55 A. Pre-save again in bank A. SOA2 low-sides the gate at 2 mA about 170 us off.
The 2 mA also pulls VDUG down through U1's OUT–GATE zener, against the 54 uA of his internal 1 MOhm SENSE–OUT path, so VBUS reads near 0 V with the bank open.
Reverse: D1 blocks current when the battery is backwards, so the banks open. DN2D-fred low: opening on overvoltage would strand regen on the bus.
LM5069-1 latches on a fault. Clear: 1-cycle cycle, then reset.
BAT spike: the absorber in blocks 10 and 15W (9 x 47 uF) takes the lead current when a bank opens fast. 0.05MΩ/5kA is the backstop. FETs and D1 are 100 V.
Polarity (rev B5): one soldered R (MA) pigtail pair, side by side, to an XT90 that only mates the pack's XT90S one way (obey the pack nut is the heaviest connector run in the robot). Nothing on the main path blocks a reversed pack: both banks' body diodes would conduct and D1 becomes the crowbar, so the keyed plug and the bring-up polarity check carry it.

Pack: 8 AWG pigtail pair to an XT90 (mates the pack XT90S), 1 BAT, 2 GND

2. Control rail and E-stop latch (no logic IC)

Block 10, Absorber: a bank trip into a bus short cuts 350–890 A (pcss/absorber_sim.py (pack 54.6 V, no pack yet), worst corners: 0.05 Ohm pulse loop (cell AC resistance), 0.5–1 uH, D1 at 71.2 V, C −20 %, trip 8.8–17.5 us, Eight N7 of R0 3 hybrids (EDH–ZCC4470P, the bus-cap part) give BAT 75 V or less (0.05 Ohm loop): 6.5 under rating, about 18 A/us, 0.2 J, 0.2 W hot. Rev B: 9 x 47 uF 100 V (10 mF) at 0.05 Ohm: 0.5 J, 0.5 W hot. Means rerun absorber_calc before motors. Measure lead 5 and pulse M at bring-up. 100k bleeds 0, 8W1 close drives about 260 A into three cans for a few us; bench it on the JK.
U2 feed T1, 22R, ESD3 in series, then 1 uF. A shorted U2 pulls the pack on VCTL; 22R holds T1 to about 1.6 A, so SMA/DSA clamps under 10.2 V (Q6 gate 20 V) and T1 250 A 10 ms, inside its 1 kA 10 ms. Bench: short VCTL to GND, confirm T1 holds, then open it.
Absorber close option (NO): DNP. If the BMS trips when it closes onto the absorber, remove the DR and T5 2R + the delayed NMOS + Vc RC: 25 A at close, clamp unchanged (absorber_sim.py).
Loop: the 4 BAS40 diodes: a 15 pin 2–3 bridge or an E8 fault can't drive RUN. F1 reaches LOOP through BAT46W + 10k (2010) with a 12 V zener on FLT_A: F1 low holds LOOP near 1.9 V, a bridged VLO puts at most 5.5 mA into FLT and 12.7 V on it (15 V recommended).
Converter: F2 2A is on the board and protects the pigtail. D7 already limits at 5.7–12.5 A: F2 backs up a shorted Q7). The 1nF has no reverse protection, so the keyed connector.
Fault: R7 33R, C5 1 uF, R8 10k, D5 SMA15A, R9 10k, D6 SMA15A, R10 4.7k, R11 10k, R12 1k, C6 2.2 uF COG, R13 1k, R14 10k, R15 1.2k, R16 10k, R17 10k, R18 10k, C7 100 nF COG, R19 499k 0.1%, R20 604k 0.1%, R21 54.1k 0.1%, R22 9.53M, C8 100 nF 50V X7R, C9 100 nF COG, R23 10R, R24 100k, R25 10R, R26 10R, C10 4.7 uF 80V EDH–ZC1K470P, C11 4.7 uF 80V EDH–ZC1K470P, C12 4.7 uF 80V EDH–ZC1K470P, C13 4.7 uF 80V EDH–ZC1K470P, C14 2.2 uF 100V X7R, C15 2.2 uF 100V X7R, C16 1 uF 80V EDH–ZC1K470P, C17 47 nF/1 kV, C18 1 uF 100V X7R, C19 220 nF COG, C20 100 nF 100V X7R, C21 4.7 uF 10V X7R, C22 4.7 uF 10V X7R, C23 4.7 uF 10V X7R, C24 4.7 uF 10V X7R, C25 4.7 uF 10V X7R, C26 4.7 uF 10V X7R, C27 4.7 uF 10V X7R, C28 4.7 uF 10V X7R, C29 4.7 uF 10V X7R, C30 4.7 uF 10V X7R, C31 100 nF 50V X7R, C32 100 nF 50V X7R, C33 100 nF 50V X7R, C34 100 nF 50V X7R, C35 100 nF 50V X7R, C36 100 nF 50V X7R, C37 100 nF 50V X7R, C38 1 uF 80V EDH–ZC1K470P, C39 100 nF 50V X7R, C40 100 nF 50V X7R, C41 100 nF 50V X7R, C42 100 nF 50V X7R, C43 100 nF 50V X7R, C44 100 nF 50V X7R, C45 100 nF 50V X7R, C46 100 nF 50V X7R, C47 4.7 uF 80V EDH–ZC1K470P, C48 4.7uF 10V X7R, C49 4.7uF 10V X7R, D1 SMA15A, D2 SMA15A, D3 SMA15A, D4 SMA15A, D5 SMA15A, D6 SMA15A, D7 SMA15A, D8 SMA15A, D9 SMA15A, D10 SMA15A, D11 SMA15A, D12 SMA15A, D13 SMA15A, D14 SMA15A, D15 SMA15A, D16 SMA15A, D17 SMA15A, D18 SMA15A, D19 SMA15A, D20 SMA15A, D21 SMA15A, D22 SMA15A, D23 SMA15A, D24 SMA15A, D25 SMA15A, D26 SMA15A, D27 SMA15A, D28 SMA15A, D29 SMA15A, D30 SMA15A, D31 SMA15A, D32 SMA15A, D33 SMA15A, D34 SMA15A, D35 SMA15A, D36 SMA15A, D37 SMA15A, D38 SMA15A, D39 SMA15A, D40 SMA15A, D41 SMA15A, D42 SMA15A, D43 SMA15A, D44 SMA15A, D45 SMA15A, D46 SMA15A, D47 SMA15A, D48 SMA15A, D49 SMA15A, D50 SMA15A, D51 SMA15A, D52 SMA15A, D53 SMA15A, D54 SMA15A, D55 SMA15A, D56 SMA15A, D57 220k, D58 SMA15A, D59 SMA15A, D60 SMA15A, D61 SMA15A, D62 SMA15A, D63 SMA15A, D64 SMA15A, D65 SMA15A, D66 SMA15A, D67 SMA15A, D68 SMA15A, D69 SMA15A, D70 SMA15A, D71 SMA15A, D72 SMA15A, D73 SMA15A, D74 SMA15A, D75 SMA15A, D76 SMA15A, D77 SMA15A, D78 SMA15A, D79 SMA15A, D80 SMA15A, D81 SMA15A, D82 SMA15A, D83 SMA15A, D84 SMA15A, D85 SMA15A, D86 SMA15A, D87 SMA15A, D88 SMA15A, D89 SMA15A, D90 SMA15A, D91 SMA15A, D92 SMA15A, D93 SMA15A, D94 SMA15A, D95 SMA15A, D96 SMA15A, D97 SMA15A, D98 SMA15A, D99 SMA15A, D100 SMA15A, D101 SMA15A, D102 SMA15A, D103 SMA15A, D104 SMA15A, D105 SMA15A, D106 SMA15A, D107 SMA15A, D108 SMA15A, D109 SMA15A, D110 SMA15A, D111 SMA15A, D112 SMA15A, D113 SMA15A, D114 SMA15A, D115 SMA15A, D116 SMA15A, D117 SMA15A, D118 SMA15A, D119 SMA15A, D120 SMA15A, D121 SMA15A, D122 SMA15A, D123 SMA15A, D124 SMA15A, D125 SMA15A, D126 SMA15A, D127 SMA15A, D128 SMA15A, D129 SMA15A, D130 SMA15A, D131 SMA15A, D132 SMA15A, D133 SMA15A, D134 SMA15A, D135 SMA15A, D136 SMA15A, D137 SMA15A, D138 SMA15A, D139 SMA15A, D140 SMA15A, D141 SMA15A, D142 SMA15A, D143 SMA15A, D144 SMA15A, D145 SMA15A, D146 SMA15A, D147 SMA15A, D148 SMA15A, D149 SMA15A, D150 SMA15A, D151 SMA15A, D152 SMA15A, D153 SMA15A, D154 SMA15A, D155 SMA15A, D156 SMA15A, D157 SMA15A, D158 SMA15A, D159 SMA15A, D160 SMA15A, D161 SMA15A, D162 SMA15A, D163 SMA15A, D164 SMA15A, D165 100k, D166 1k, D167 SMA15A, D168 SMA15A, D169 SMA15A, D170 SMA15A, D171 SMA15A, D172 SMA15A, D173 SMA15A, D174 SMA15A, D175 SMA15A, D176 SMA15A, D177 SMA15A, D178 SMA15A, D179 SMA15A, D180 SMA15A, D181 SMA15A, D182 SMA15A, D183 SMA15A, D184 SMA15A, D185 SMA15A, D186 SMA15A,